רשתות תקשורת עבור מרכזי נתונים ובינה מלאכותית 361.2.2460

סמסטר א׳ – תשפ״ז – 2026

Homework 1 (ver 2)

Topic: Fat-Tree Network Topology Generation and Analysis

Background and Objective

In modern data center networks, **fat-tree topologies** are widely used due to their scalability, fault tolerance, and high bisection bandwidth. This assignment aims to familiarize you with the structural properties of fat-tree architectures and provide hands-on experience in programmatically generating and analyzing such topologies.
Through this exercise, you will design and implement a topology generator, introduce network perturbations (e.g., link failures), and evaluate how these affect selected performance or structural metrics. The goal is to deepen your understanding of hierarchical network design principles and their implications for robustness and scalability.

1. Submission Requirements

Submit your work as a **link to a GitHub repository** containing:

- The full **source code** of your implementation.
- A README.md file that clearly explains your approach, provides **instructions for running the code**, and summarizes your **results**.

2. Topology Design

Implement a **fat-tree network topology** consistent with the design presented in paper [1]. You may refer to additional academic sources as needed.
Your implementation should realize a **three-tier hierarchical structure**.

3. Input Specification

Define and document the **input parameters** required by your topology-generation program.
Your code should be flexible enough to accommodate different network scales or configurations.

4. Failure Modeling

Extend your implementation to accept an **optional command-line parameter or a config file** specifying the **percentage of link failures** in the network. This parameter should supplement the standard inputs required for topology generation, allowing you to analyze network behavior under failure conditions.

5. Metric Analysis and Visualization

Perform a quantitative analysis of how a **non-trivial network metric** (that you choos) varies as a function of one of your input parameters. Choose a metric and experimental design that address a meaningful or insightful question regarding the network's structure or performance.

Examples include (but are not limited to, you can think of your own):

- Average path length (or number of paths) as a function of link failure probability (advanced)
- Number of switches with multiple failed links (at a 1% failure rate) as a function of switch port count (intermediate)
- Number of hosts supported by the network as a function of switch port count (basic)

Visualize your results using appropriate plots or charts.

6. Documentation of Results

For each plot or analysis you produce, include an **illustrative example** in your README.md file. Each example should be accompanied by:

- A short description of the setup, the experiment, and its parameters.
- A discussion of the observed results, an explanation and their significance.

7. Submission deadlines

- **22/10/26 08:00 –** A link to the GitHub with an initial code and a README.md file – 20%
- **25/10/26 12:00** – Full homework – 80%
- **On 22/10/26 and 25/10/26, 1/11/26 we will discuss some of the results in the class, be prepare to present your results.**

AI usage: Recommended, with a clear reference. You should know and understand everything you submit, and it is your full responsibility for it.

Good luck and enjoy!

[1] M. Al-Fares, A. Loukissas, and A. Vahdat, "A scalable, commodity data center network architecture," SIGCOMM Comput. Commun. Rev., vol. 38, no. 4, pp. 63–74, Oct. 2008, doi: 10.1145/1402946.1402967.